

# Movie Recommendation

Solution Engineering Python

[https://github.com/AntonyShaps/solution\\_eng\\_movies](https://github.com/AntonyShaps/solution_eng_movies)





A blurred background image showing people in a movie theater. In the foreground, a hand holds a large black bucket of popcorn. In the background, other people are visible, some holding popcorn, creating a cinematic atmosphere.

## MODELS

SVD

Neural Collaborative Filtering

K-Means/KNN

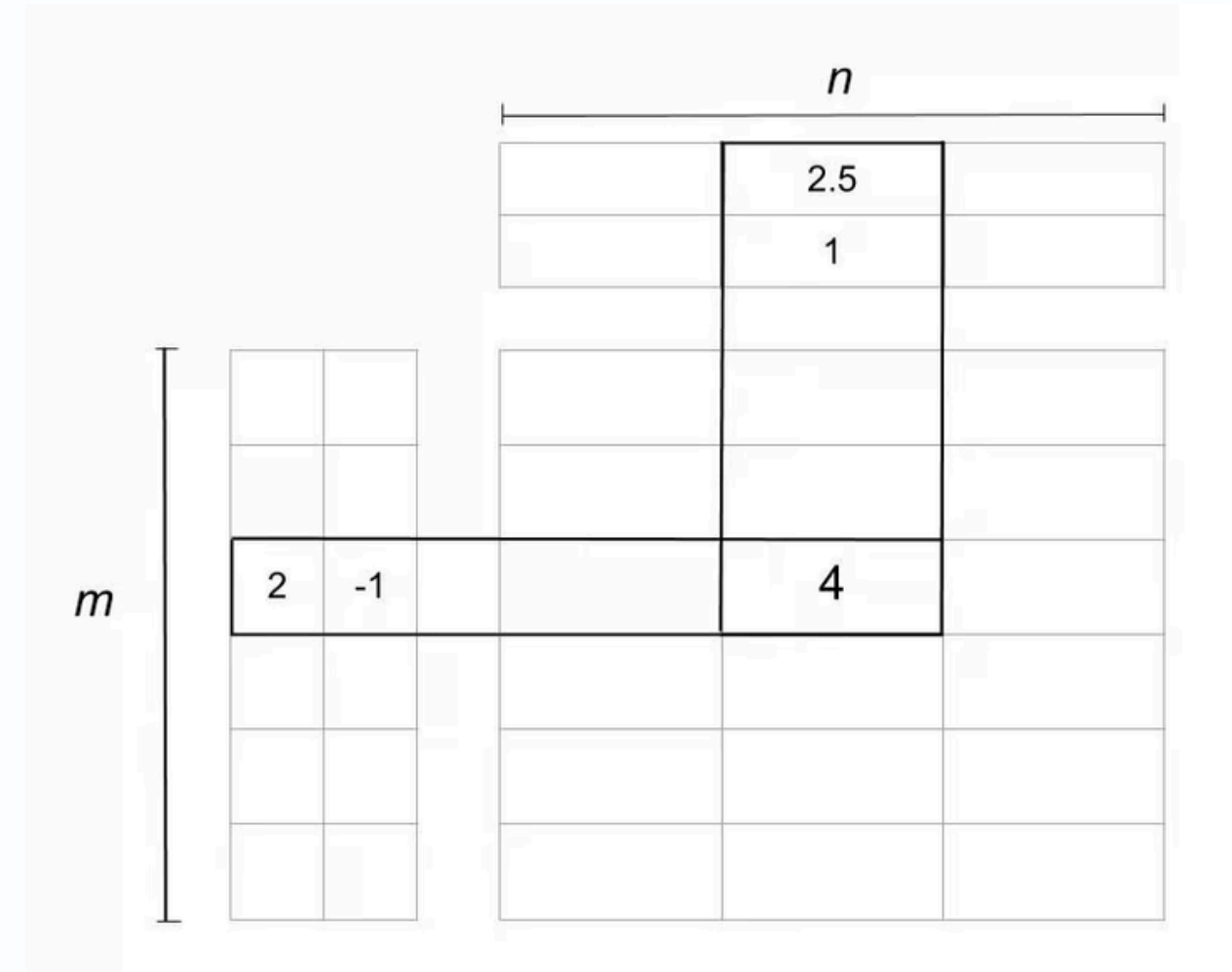
basic recommendations



# Singular Value Decomposition

Collaborative Filtering by Dimensionality Reduction

- Dimensionality reduction by Matrix Factorization
- The Users and Movies are represented individually
- The one on the left is the user matrix with  $m$  users, and the one on top is the item matrix with  $n$  items. The rating 4 is reduced or factorized into:
  - A user vector  $(2, -1)$
  - A movies vector  $(2.5, 1)$
- The characteristics of User/Movies are represented by their vectors which are called Latent Factors



# SVD Experiment

- Full Validation Set to train the model
- Hyperparameter tuning with RandomizedSearchCV
- Default RMSE (0.8124) and MAE (0.6158)

```
model = SVD()
model.fit(trainset)
predictions = model.test(testset)
rmse_default = accuracy.rmse(predictions)
mae_default = accuracy.mae(predictions)
```

RMSE: 0.8124  
MAE: 0.6158

## Hyperparameter Tuning

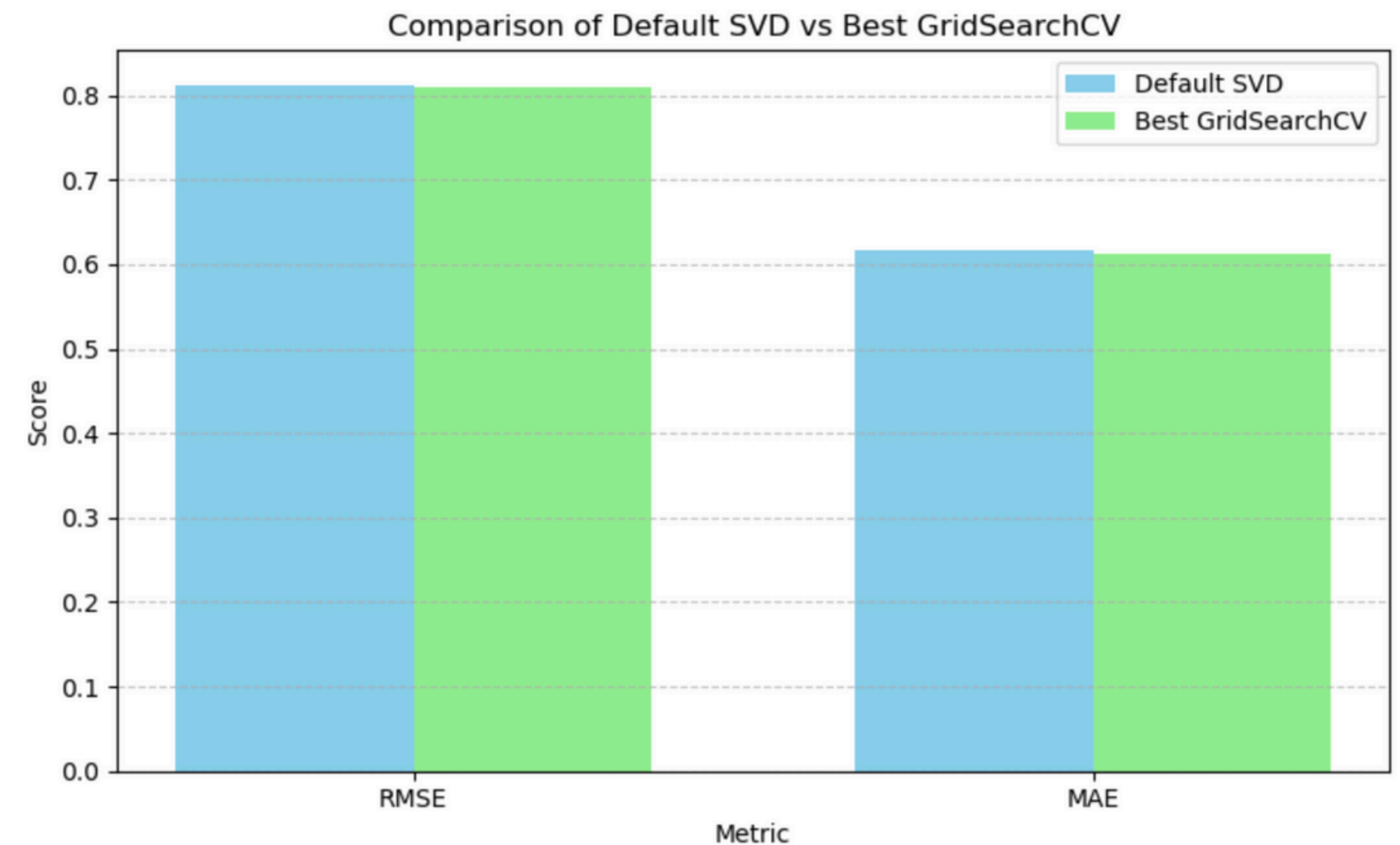
- n\_factors: Factors(default: 100) (high factors -> overfitting)
- n\_epochs: epochs
- lr\_all: Learning Rate (default:0.005)
- reg\_all: regularization term (default:0.02)
- random\_state: seed

```
param_grid = {
    'n_factors': [50, 100],
    'n_epochs': [20, 30, 50],
    'lr_all': [0.002, 0.005],
    'reg_all': [0.02, 0.05],
    'random_state': [42]
}

gs = RandomizedSearchCV(SVD, param_grid, n_iter=5, measures=['rmse'], cv=3, n_jobs=-1)
gs.fit(data)

print(gs.best_score['rmse'])
print(gs.best_params['rmse'])
```

0.8125777894627223  
{'n\_factors': 50, 'n\_epochs': 30, 'lr\_all': 0.005, 'reg\_all': 0.02, 'random\_state': 42}



There is some improvement in both the metrics but not by a lot.



# SVDpp Experiment

- Full Validation Set to train the model
- Hyperparameter tuning with RandomizedSearchCV
- Default RMSE (0.8057) and MAE (0.6075)

```
model = SVDpp()
model.fit(trainset)
predictions = model.test(testset)
rmse_default = accuracy.rmse(predictions)
mae_default = accuracy.mae(predictions)
```

RMSE: 0.8057  
MAE: 0.6075

## Hyperparameter Tuning

- n\_factors: Factors (default: 100) (high factors -> overfitting)
- n\_epochs: epochs
- lr\_all: Learning Rate (default: 0.005)
- reg\_all: regularization term (default: 0.02)
- random\_state: seed

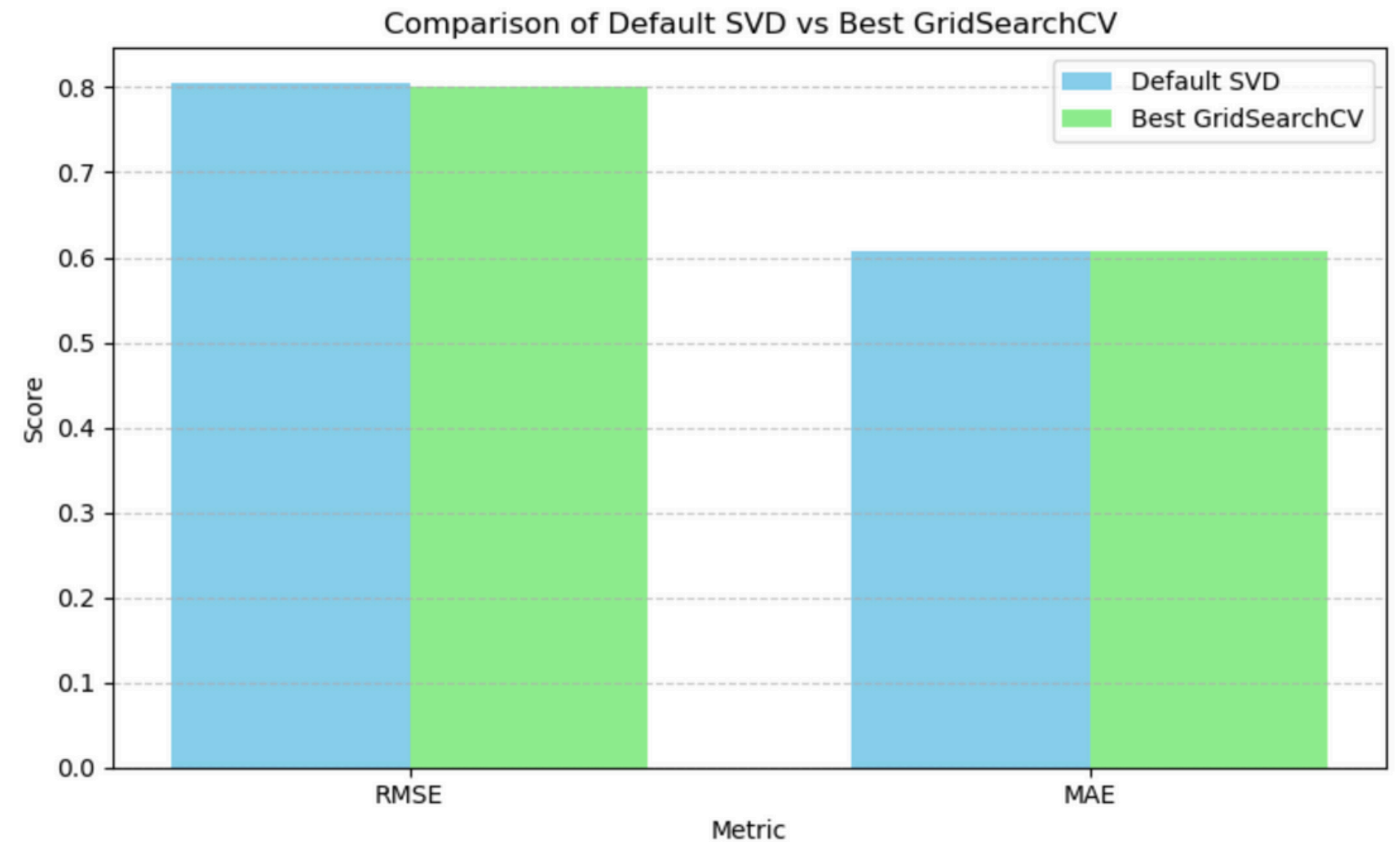
```
param_grid = {
    'n_factors': [50, 100],
    'cache_ratings': [True, False],
    'n_epochs': [20, 30, 50],
    'lr_all': [0.002, 0.005],
    'reg_all': [0.02, 0.05],
    'random_state': [42]
}

gs = RandomizedSearchCV(SVDpp, param_grid, n_iter=5, measures=['rmse'], cv=3, n_jobs=-1)
gs.fit(data)

print(gs.best_score['rmse'])
print(gs.best_params['rmse'])
```

0.8027741372024667

{'n\_factors': 100, 'cache\_ratings': False, 'n\_epochs': 30, 'lr\_all': 0.005, 'reg\_all': 0.05, 'random\_state': 42}



There is some improvement in both the metrics but not by a lot.



# SVD Experiment Conclusions

Need more Power

- SVDpp has better RMSE and MAE as compared to SVD
- The training time for SVDpp was 407 minutes as compared to 30 minutes for SVD on the validation set (3.5 million rows)
- One probable cause could be how SVDpp works internally as it adds an implicit feedback as well i.e. takes into consideration what the current user has rated as well
- Tried different cloud providers to speed up the training
  - Deepnote
  - Google Colab
  - Amazon EC2
  - AWS SageMaker Studio Lab
- However, the memory requirements were not fulfilled due to free tier
- Also, retraining on a cached model with input user ratings also did not work with following approaches
  - Cosine Similarity
  - Euclidean distance
- Only retraining with the full dataset (subset of 50k rows) again based on user ratings seemed to work



# SVD Recommender

- Downsized Ratings to predict movies in reasonable time (from fit train to predict)
  - 10k – 0.35 s
  - 20k – 0.48 s
  - 50k – 1.01 s
- Model configured with best Hyperparameters
- 5 ratings (1-5 stars) as input to generate 10 recommendations



### Your Rating



☒ ☒

Show Your Ratings

|   | movieid | title                              | rating |
|---|---------|------------------------------------|--------|
| 0 | 318     | The Shawshank Redemption           | 5      |
| 1 | 593     | The Silence of the Lambs           | 4      |
| 2 | 2,571   | The Matrix                         | 5      |
| 3 | 527     | Schindler's List                   | 5      |
| 4 | 260     | Star Wars: Episode IV - A New Hope | 5      |

Select Recommendation Model

SVD

### Recommendations (🕒 0.35 sec)



Select Recommendation Model

SVD

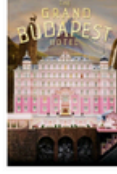
### Recommendations (🕒 0.48 sec)



Select Recommendation Model

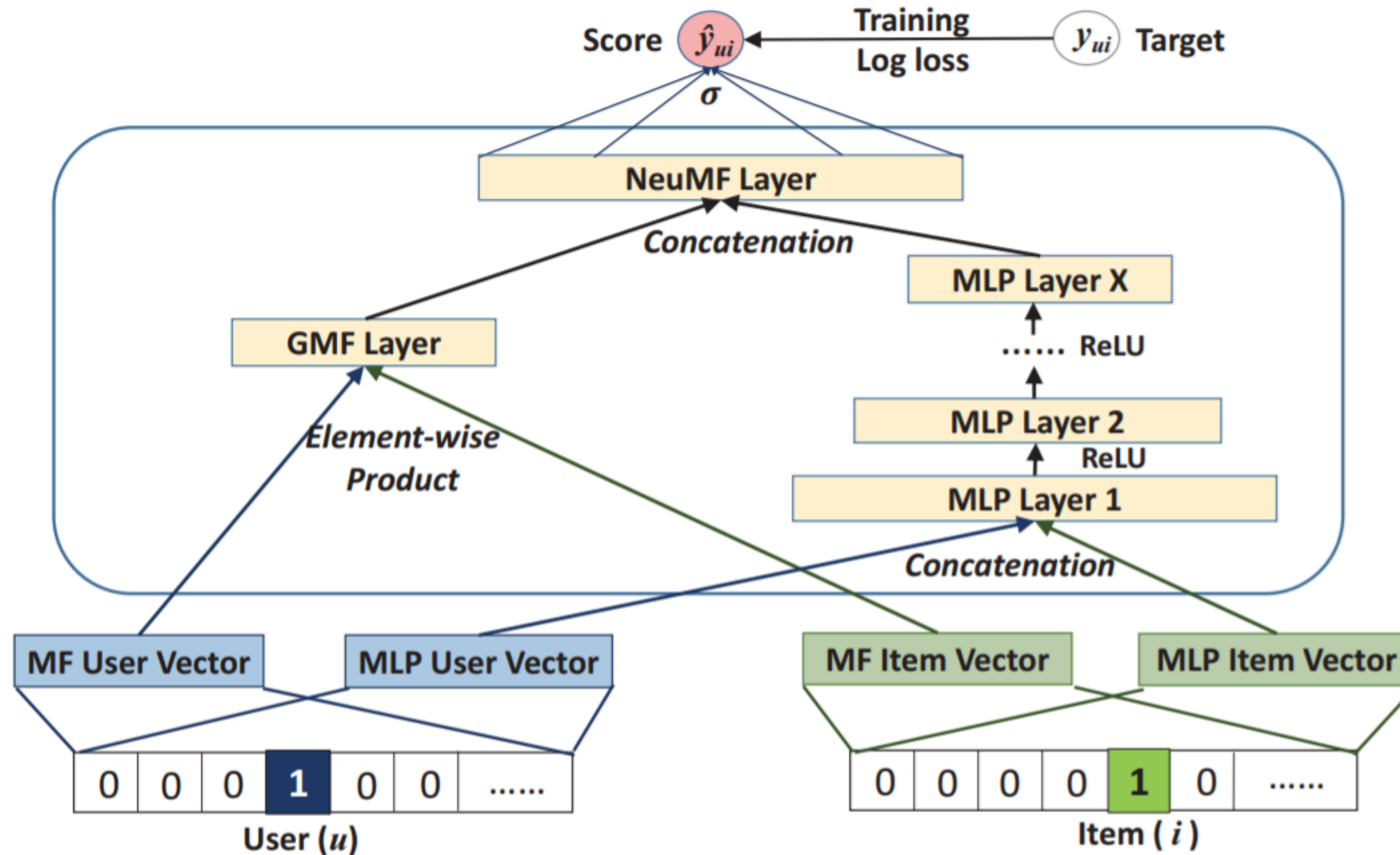
SVD

### Recommendations (🕒 1.01 sec)



# Neural Collaborative Filtering

NeuMF is a part of RecBole library





# NCF: Model 4

Trained for 6 hours (14 epoch)

---

```
)
(predict_layer): Linear(in_features=128, out_features=1, bias=True)
(sigmoid): Sigmoid()
(loss): BCEWithLogitsLoss()
)
Trainable parameters: 28388417
Sat 31 May 2025 11:08:41 INFO FLOPs: 24960.0
Sat 31 May 2025 11:29:10 INFO epoch 0 training [time: 1227.70s, train loss: 6181.0007]
Sat 31 May 2025 11:48:58 INFO epoch 1 training [time: 1188.95s, train loss: 4719.6027]
Sat 31 May 2025 12:08:48 INFO epoch 2 training [time: 1189.37s, train loss: 4335.5745]
Sat 31 May 2025 12:28:43 INFO epoch 3 training [time: 1195.60s, train loss: 4155.9738]
Sat 31 May 2025 12:48:39 INFO epoch 4 training [time: 1195.58s, train loss: 4046.8818]
Sat 31 May 2025 12:50:59 INFO epoch 4 evaluating [time: 139.96s, valid_score: 0.945300]
Sat 31 May 2025 12:50:59 INFO valid result:
recall@10 : 0.8936 ndcg@10 : 0.9453 mrr@10 : 0.9604
Sat 31 May 2025 12:51:00 INFO Saving current: saved_models_ml25m_ranking/NeuMF-
May-31-2025_11-08-41.pth
Sat 31 May 2025 13:10:27 INFO epoch 5 training [time: 1167.62s, train loss: 3975.3278]
Sat 31 May 2025 13:30:05 INFO epoch 6 training [time: 1177.90s, train loss: 3917.1133]
Sat 31 May 2025 13:49:30 INFO epoch 7 training [time: 1164.77s, train loss: 3875.4026]
Sat 31 May 2025 14:08:56 INFO epoch 8 training [time: 1166.36s, train loss: 3823.6742]
Sat 31 May 2025 14:28:23 INFO epoch 9 training [time: 1166.46s, train loss: 3789.0804]
Sat 31 May 2025 14:30:37 INFO epoch 9 evaluating [time: 134.81s, valid_score: 0.947300]
Sat 31 May 2025 14:30:37 INFO valid result:
recall@10 : 0.8941 ndcg@10 : 0.9473 mrr@10 : 0.9628
Sat 31 May 2025 14:30:38 INFO Saving current: saved_models_ml25m_ranking/NeuMF-
May-31-2025_11-08-41.pth
Sat 31 May 2025 14:50:27 INFO epoch 10 training [time: 1189.17s, train loss: 3748.0825]
Sat 31 May 2025 15:10:21 INFO epoch 11 training [time: 1194.37s, train loss: 3719.8865]
Sat 31 May 2025 15:30:16 INFO epoch 12 training [time: 1194.75s, train loss: 3689.0542]
Sat 31 May 2025 15:50:11 INFO epoch 13 training [time: 1194.93s, train loss: 3669.2649]
```

# NCF: Model 5

Trained for 18 hours (50 epoch)

```
)
(predict_layer): Linear(in_features=128, out_features=1, bias=True)
(sigmoid): Sigmoid()
(loss): BCEWithLogitsLoss()
)
Trainable parameters: 28388417
Sat 31 May 2025 23:14:35 INFO FLOPs: 24960.0
Sat 31 May 2025 23:35:00 INFO epoch 0 training [time: 1224.11s, train loss: 6181.0007]
Sat 31 May 2025 23:55:22 INFO epoch 1 training [time: 1221.76s, train loss: 4719.6027]
Sun 01 Jun 2025 00:15:45 INFO epoch 2 training [time: 1223.25s, train loss: 4335.5745]
Sun 01 Jun 2025 00:36:15 INFO epoch 3 training [time: 1230.27s, train loss: 4155.9738]
Sun 01 Jun 2025 00:56:45 INFO epoch 4 training [time: 1229.79s, train loss: 4046.8818]
Sun 01 Jun 2025 01:01:10 INFO epoch 4 evaluating [time: 264.53s, valid_score: 0.737700]
Sun 01 Jun 2025 01:01:10 INFO valid result:
.....
.....
recall@10 : 0.7928      ndcg@10 : 0.7522      mrr@10 : 0.7986
Sun 01 Jun 2025 15:37:11 INFO epoch 45 training [time: 1242.37s, train loss: 3474.9558]
Sun 01 Jun 2025 15:57:48 INFO epoch 46 training [time: 1237.09s, train loss: 3463.6348]
Sun 01 Jun 2025 16:18:23 INFO epoch 47 training [time: 1234.79s, train loss: 3462.0604]
Sun 01 Jun 2025 16:38:58 INFO epoch 48 training [time: 1235.07s, train loss: 3457.9282]
Sun 01 Jun 2025 16:59:42 INFO epoch 49 training [time: 1244.08s, train loss: 3461.1759]
Sun 01 Jun 2025 17:04:05 INFO epoch 49 evaluating [time: 263.60s, valid_score: 0.750900]
Sun 01 Jun 2025 17:04:05 INFO valid result:
recall@10 : 0.7913      ndcg@10 : 0.7509      mrr@10 : 0.7989
```



# NCF user activity

Recorded ratingsby User '345':

|     | userId | movieId | rating | timestamp  | title                                             | genres                                 |
|-----|--------|---------|--------|------------|---------------------------------------------------|----------------------------------------|
| 0   | 345    | 16      | 5.0    | 1046039132 | Casino (1995)                                     | Crime Drama                            |
| 1   | 345    | 29      | 5.0    | 1013911784 | City of Lost Children, The (Cité des enfants p... | Adventure Drama Fantasy Mystery Sci-Fi |
| 2   | 345    | 47      | 5.0    | 984709731  | Seven (a.k.a. Se7en) (1995)                       | Mystery Thriller                       |
| 3   | 345    | 50      | 5.0    | 984709701  | Usual Suspects, The (1995)                        | Crime Mystery Thriller                 |
| 4   | 345    | 70      | 3.0    | 1013922558 | From Dusk Till Dawn (1996)                        | Action Comedy Horror Thriller          |
| ... | ...    | ...     | ...    | ...        | ...                                               | ...                                    |
| 335 | 345    | 7939    | 5.0    | 1088037703 | Through a Glass Darkly (Såsom i en spegel) (1961) | Drama                                  |
| 336 | 345    | 8405    | 4.5    | 1088037595 | Hour of the Wolf (Vargtimmen) (1968)              | Drama Horror                           |
| 337 | 345    | 8507    | 4.5    | 1088037470 | Freaks (1932)                                     | Crime Drama Horror                     |
| 338 | 345    | 37733   | 3.0    | 1145418475 | History of Violence, A (2005)                     | Action Crime Drama Thriller            |
| 339 | 345    | 44195   | 2.5    | 1145418436 | Thank You for Smoking (2006)                      | Comedy Drama                           |

340 rows x 6 columns

# NCF comparison

## Model 4

Top 5 Predicted Ratings for User '345':



| Movie                      | item_id | predicted_rating |
|----------------------------|---------|------------------|
| Fight Club (1999)          | 2959    | 0.999828         |
| Almost Famous (2000)       | 3897    | 0.999680         |
| American History X (1998)  | 2329    | 0.999637         |
| American Pie (1999)        | 2706    | 0.999489         |
| The Royal Tenenbaum (2001) | 4979    | 0.999485         |

## Model 5

Top 5 Predicted Ratings for User '345':



| Movie                                                | item_id | predicted_rating |
|------------------------------------------------------|---------|------------------|
| Fight Club (1999)                                    | 2959    | 0.999935         |
| American History X (1998)                            | 2329    | 0.999676         |
| Lord of the Rings: The Fellowship of the Ring (2001) | 4993    | 0.999579         |
| Almost Famous (2000)                                 | 3897    | 0.999574         |
| Shrek (2001)                                         | 4306    | 0.999466         |



# NCF comparison

## Model 4

250 highly recommended movies (>0.99)

Top 5 Predicted Ratings for User '345':



| Movie                      | item_id | predicted_rating |
|----------------------------|---------|------------------|
| Fight Club (1999)          | 2959    | 0.999828         |
| Almost Famous (2000)       | 3897    | 0.999680         |
| American History X (1998)  | 2329    | 0.999637         |
| American Pie (1999)        | 2706    | 0.999489         |
| The Royal Tenenbaum (2001) | 4979    | 0.999485         |

## Model 5

171 highly recommended movies (>0.99)

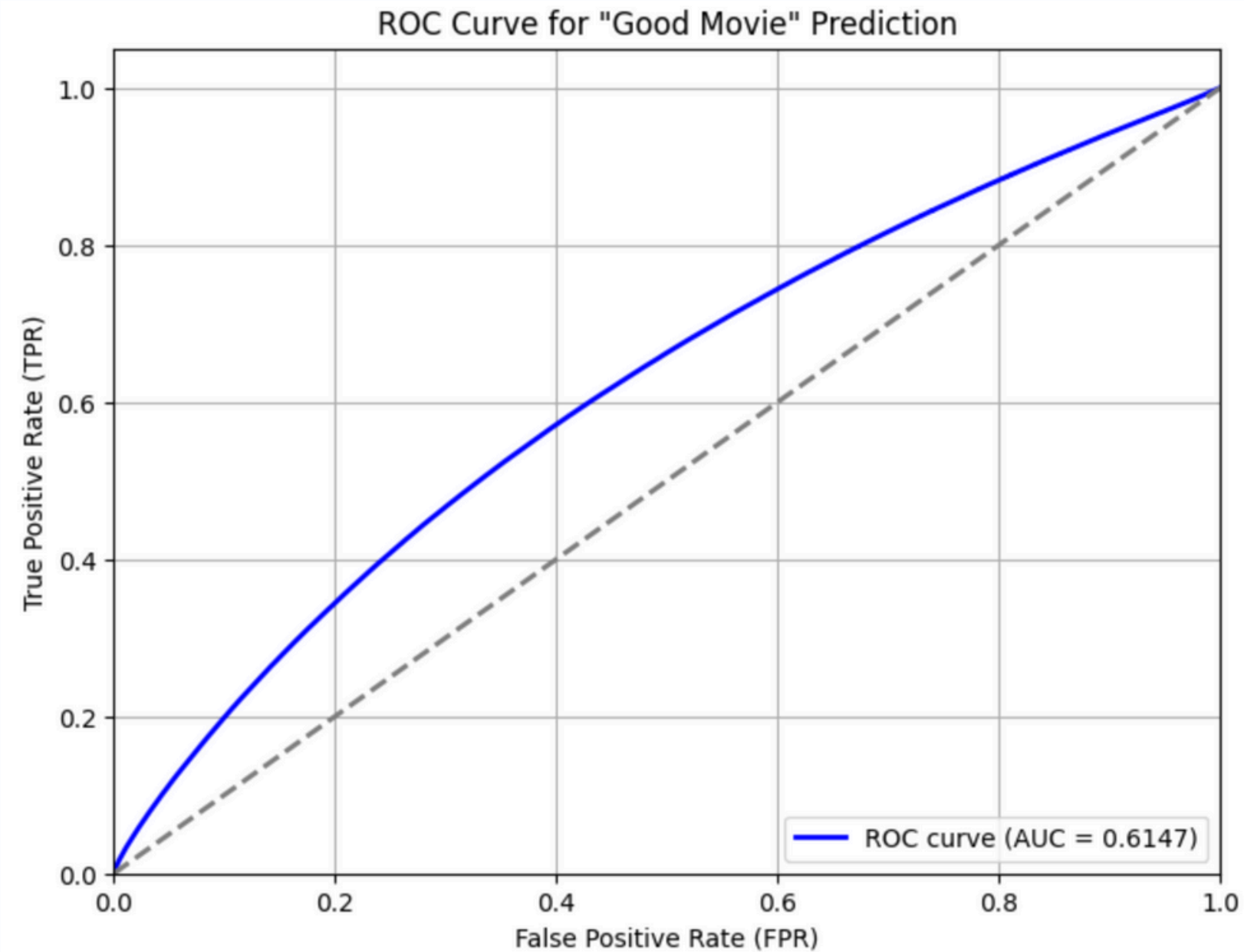
Top 5 Predicted Ratings for User '345':



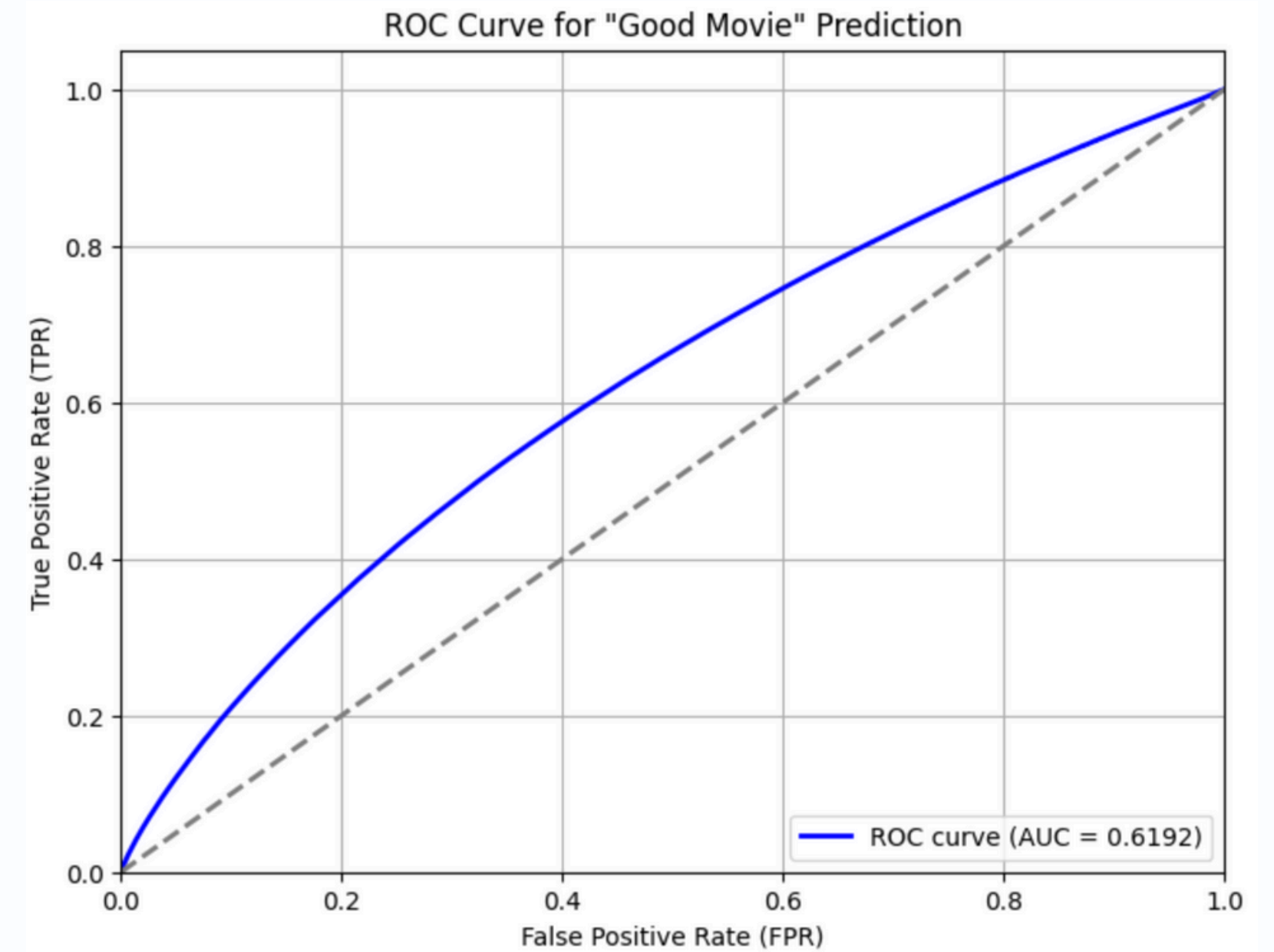
| Movie                                                | item_id | predicted_rating |
|------------------------------------------------------|---------|------------------|
| Fight Club (1999)                                    | 2959    | 0.999935         |
| American History X (1998)                            | 2329    | 0.999676         |
| Lord of the Rings: The Fellowship of the Ring (2001) | 4993    | 0.999579         |
| Almost Famous (2000)                                 | 3897    | 0.999574         |
| Shrek (2001)                                         | 4306    | 0.999466         |

# NCF comparison

## Model 4



## Model 5



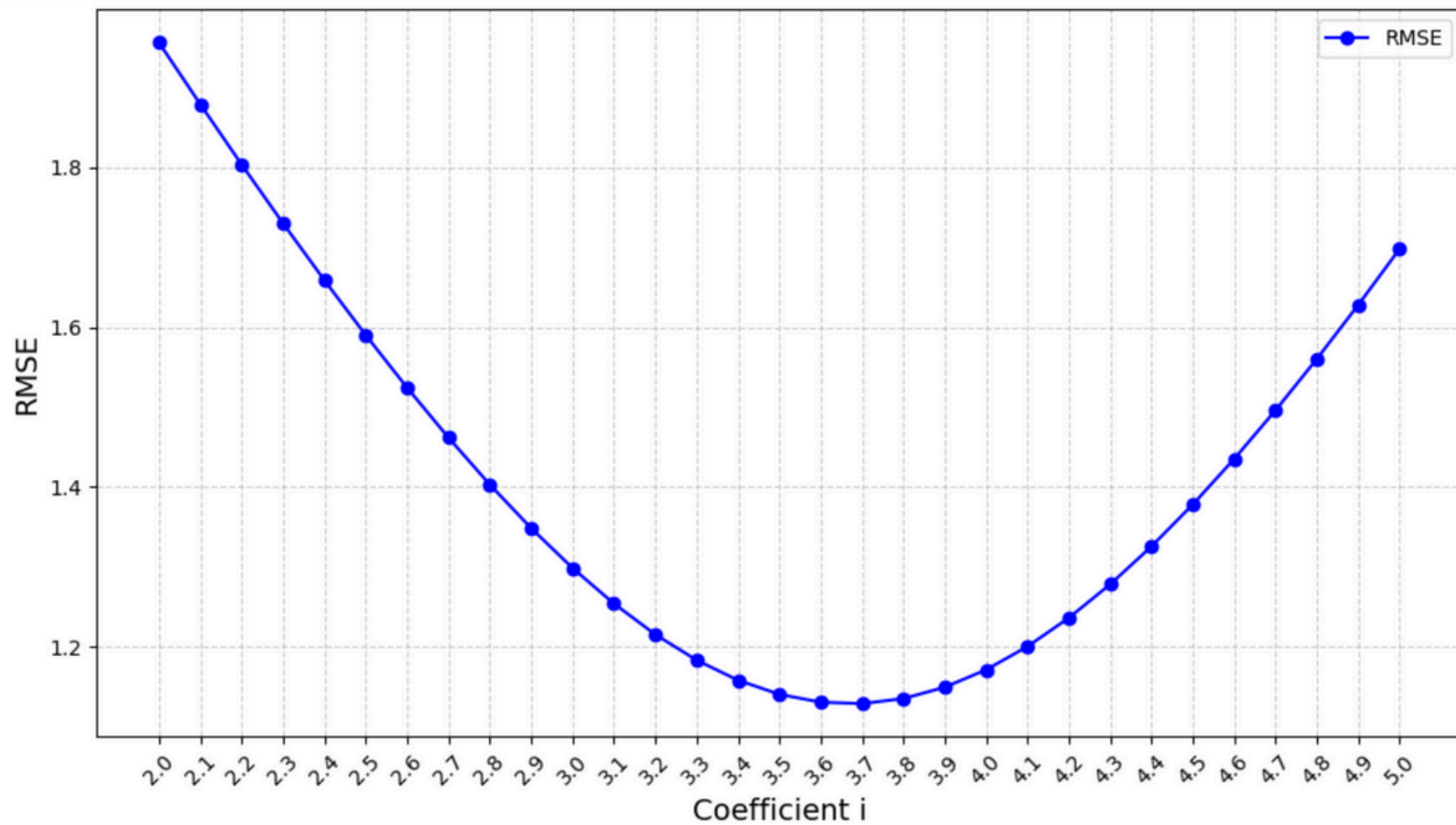
\* – “Good movie” is a movie with rating  $\geq 4.5$



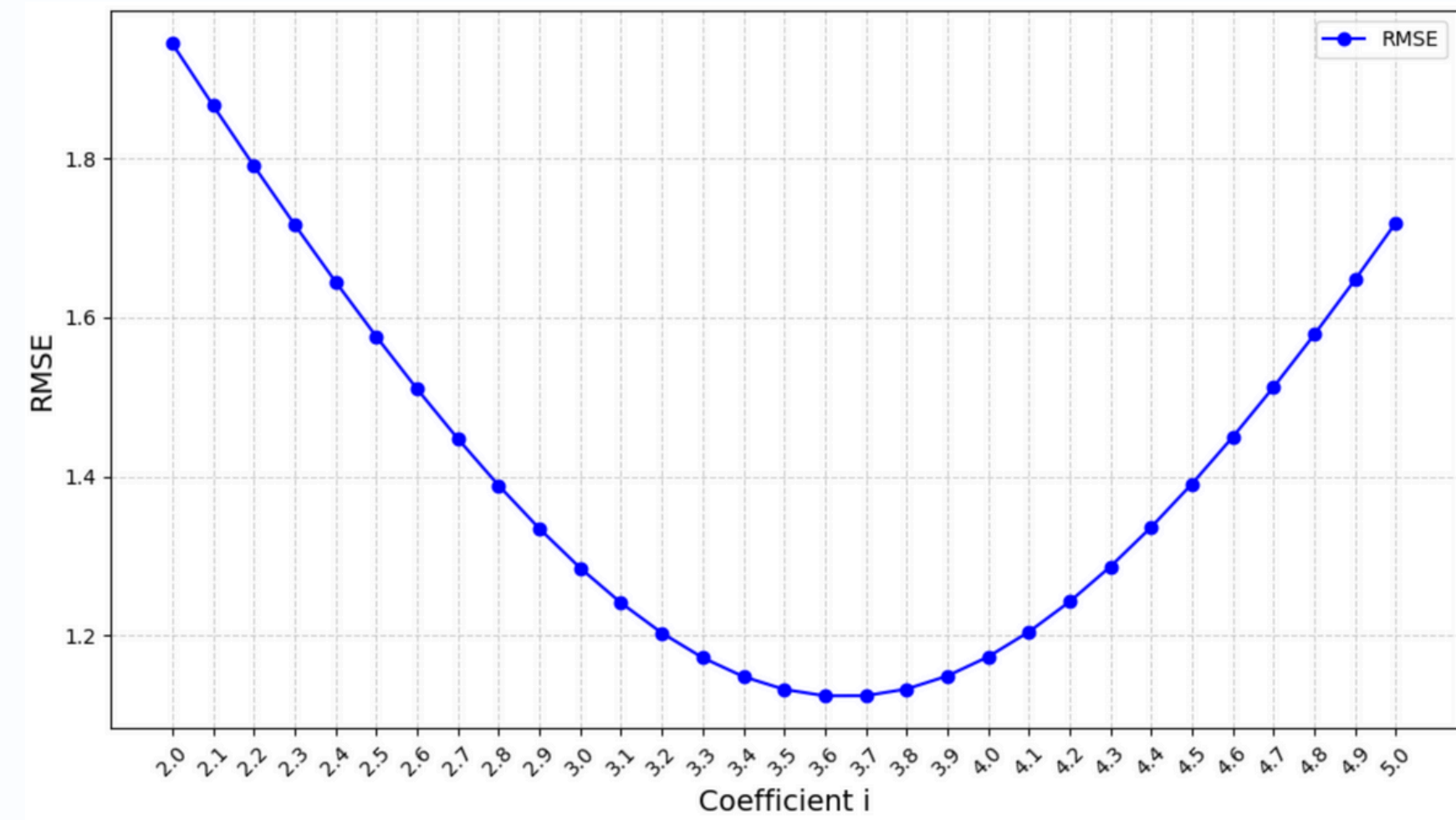
# NCF rating coefficient

How far was the predicted rating (calculated with different coefficients)?

## Model 4

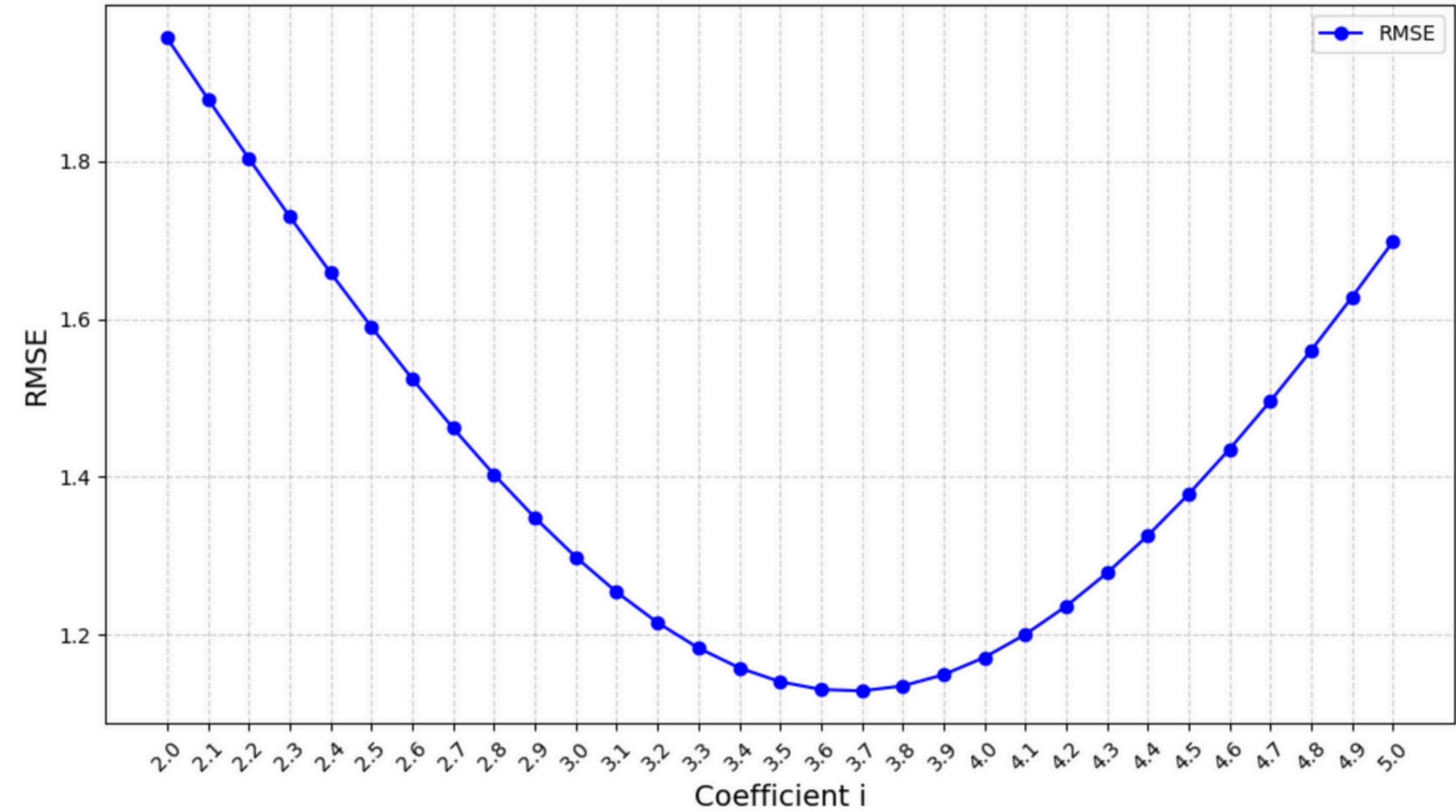


## Model 5

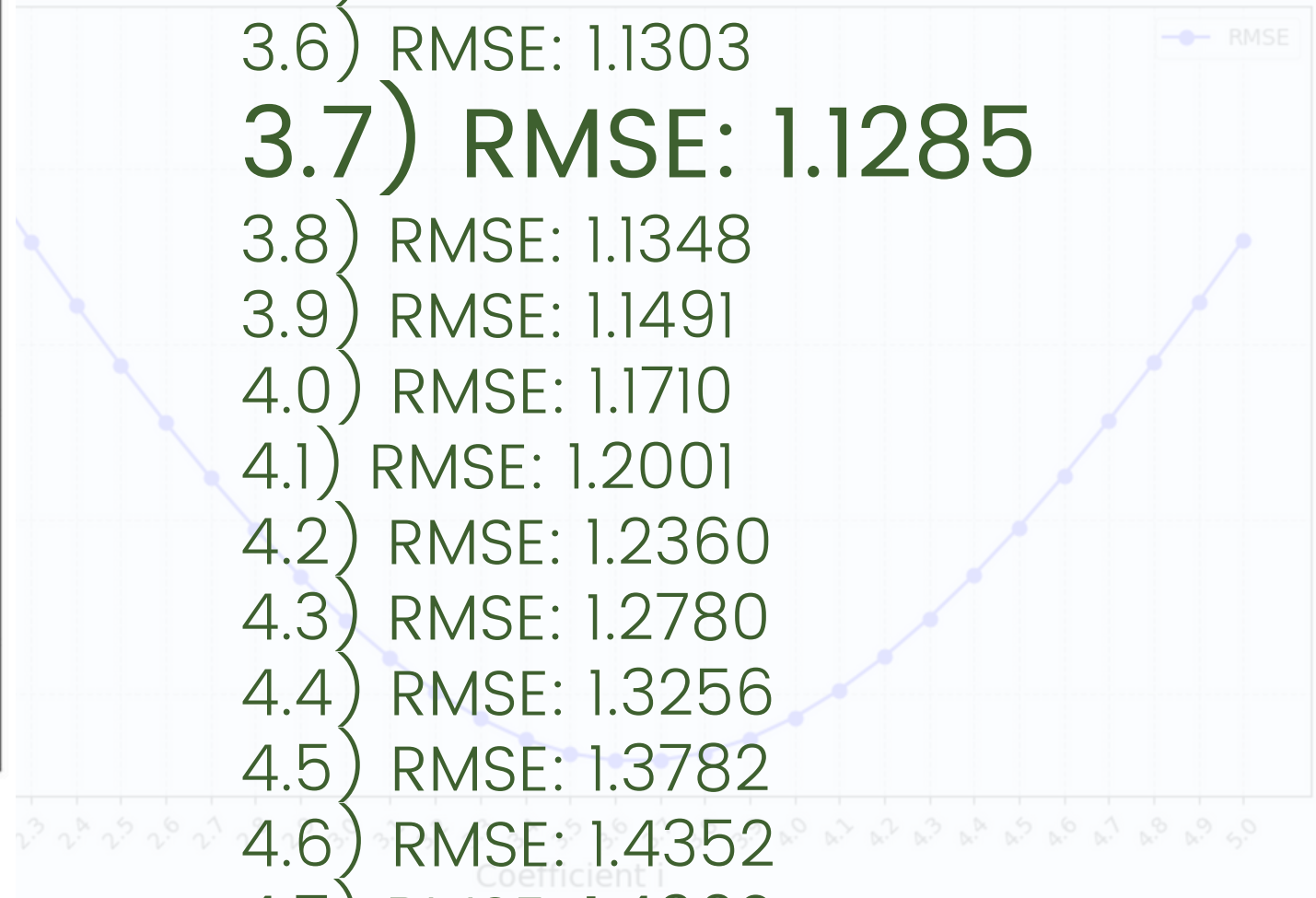


# NCF rating coefficient

## Model 4



## Model 5



- 3.0) RMSE: 1.2983
- 3.1) RMSE: 1.2538
- 3.2) RMSE: 1.2151
- 3.3) RMSE: 1.1830
- 3.4) RMSE: 1.1578
- 3.5) RMSE: 1.1401
- 3.6) RMSE: 1.1303
- 3.7) RMSE: 1.1285**
- 3.8) RMSE: 1.1348
- 3.9) RMSE: 1.1491
- 4.0) RMSE: 1.1710
- 4.1) RMSE: 1.2001
- 4.2) RMSE: 1.2360
- 4.3) RMSE: 1.2780
- 4.4) RMSE: 1.3256
- 4.5) RMSE: 1.3782
- 4.6) RMSE: 1.4352
- 4.7) RMSE: 1.4962
- 4.8) RMSE: 1.5607
- 4.9) RMSE: 1.6282
- 5.0) RMSE: 1.6984



# NCF rating coefficient

Model 4

3.0) RMSE: 1.2850

3.1) RMSE: 1.2411

3.2) RMSE: 1.2033

3.3) RMSE: 1.1723

3.4) RMSE: 1.1485

3.5) RMSE: 1.1324

**3.6) RMSE: 1.1243**

3.7) RMSE: 1.1245

3.8) RMSE: 1.1328

3.9) RMSE: 1.1492

4.0) RMSE: 1.1733

4.1) RMSE: 1.2047

4.2) RMSE: 1.2427

4.3) RMSE: 1.2868

4.4) RMSE: 1.3364

4.5) RMSE: 1.3909

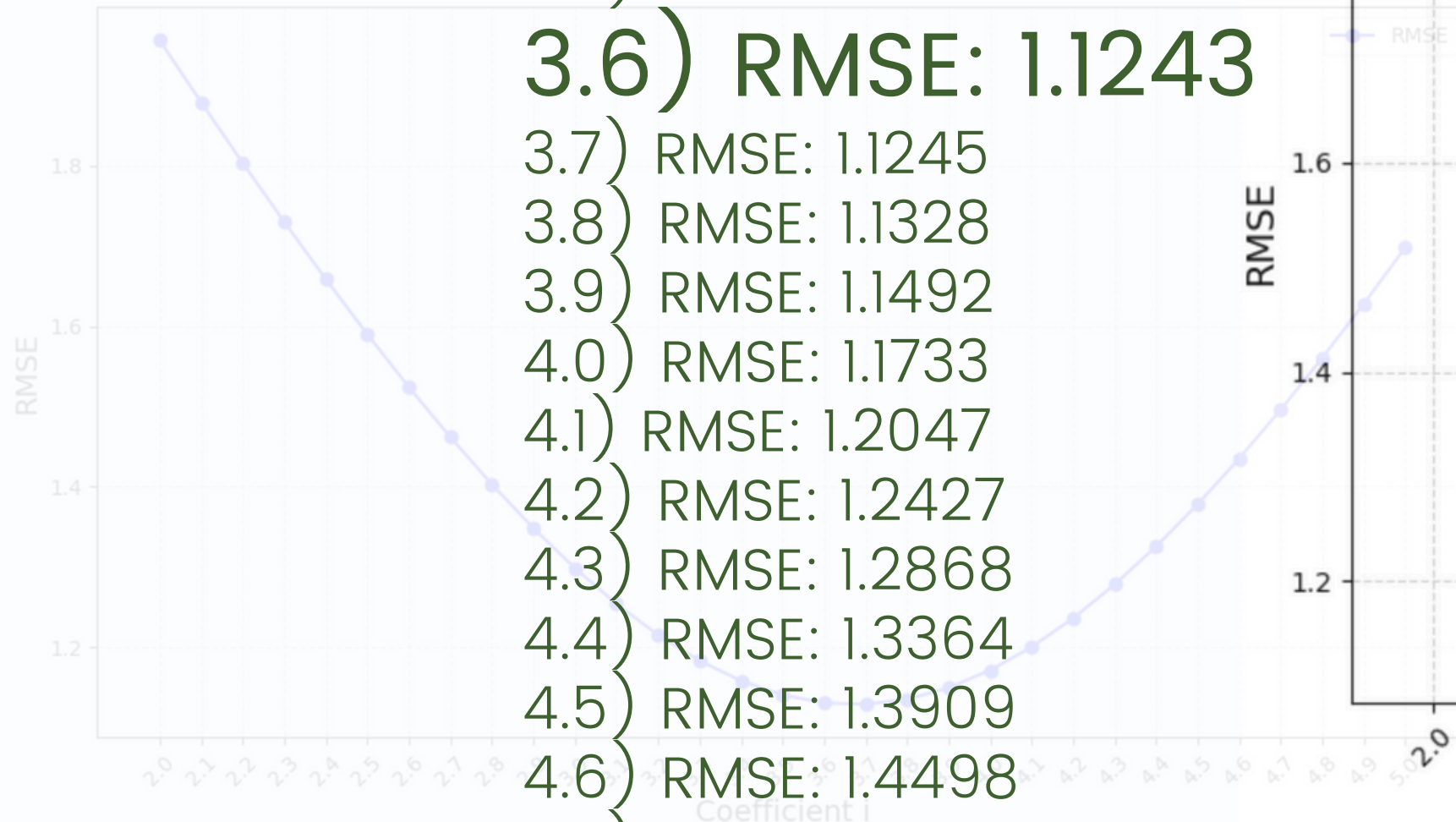
4.6) RMSE: 1.4498

4.7) RMSE: 1.5125

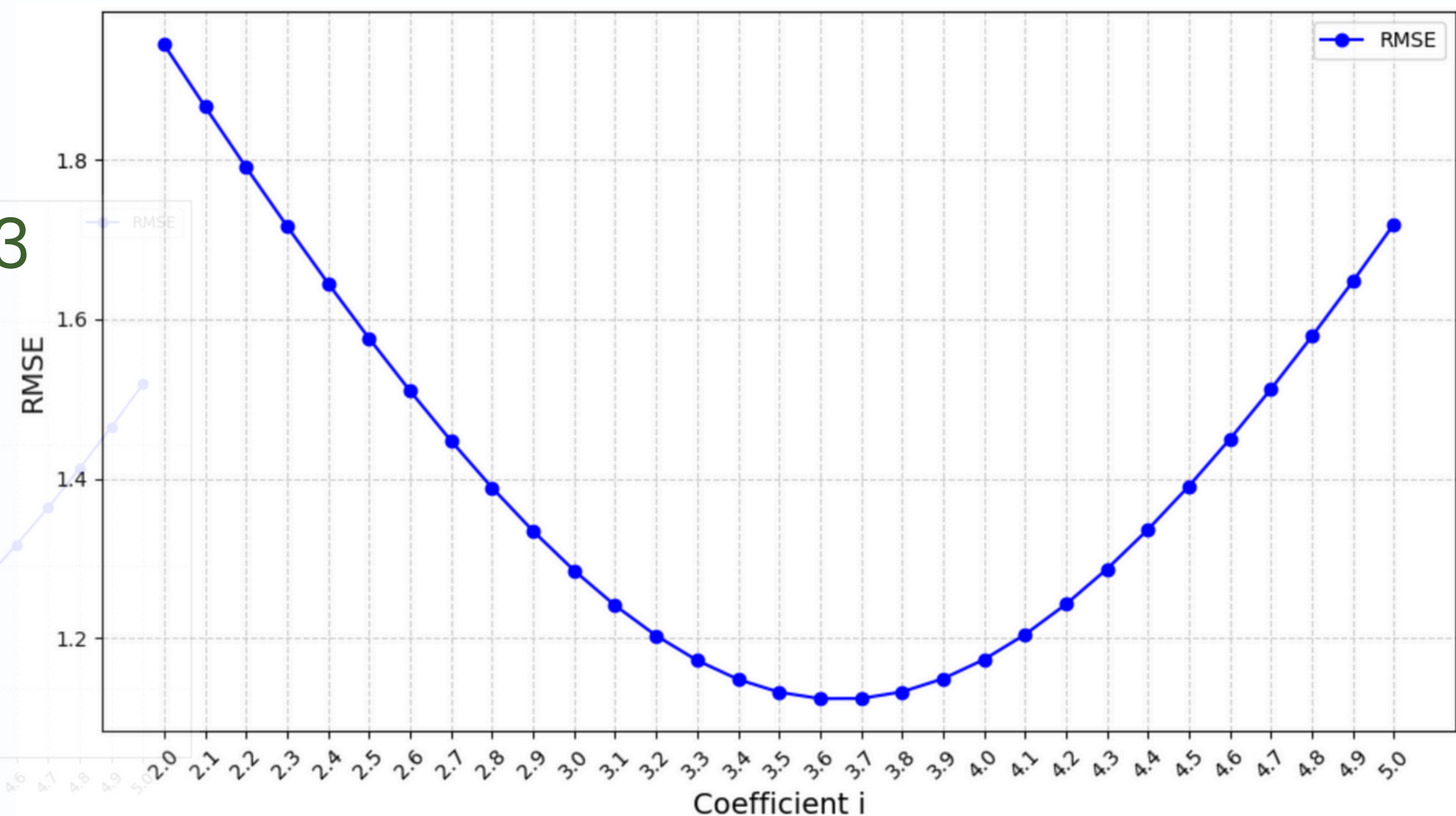
4.8) RMSE: 1.5786

4.9) RMSE: 1.6476

5.0) RMSE: 1.7193



Model 5

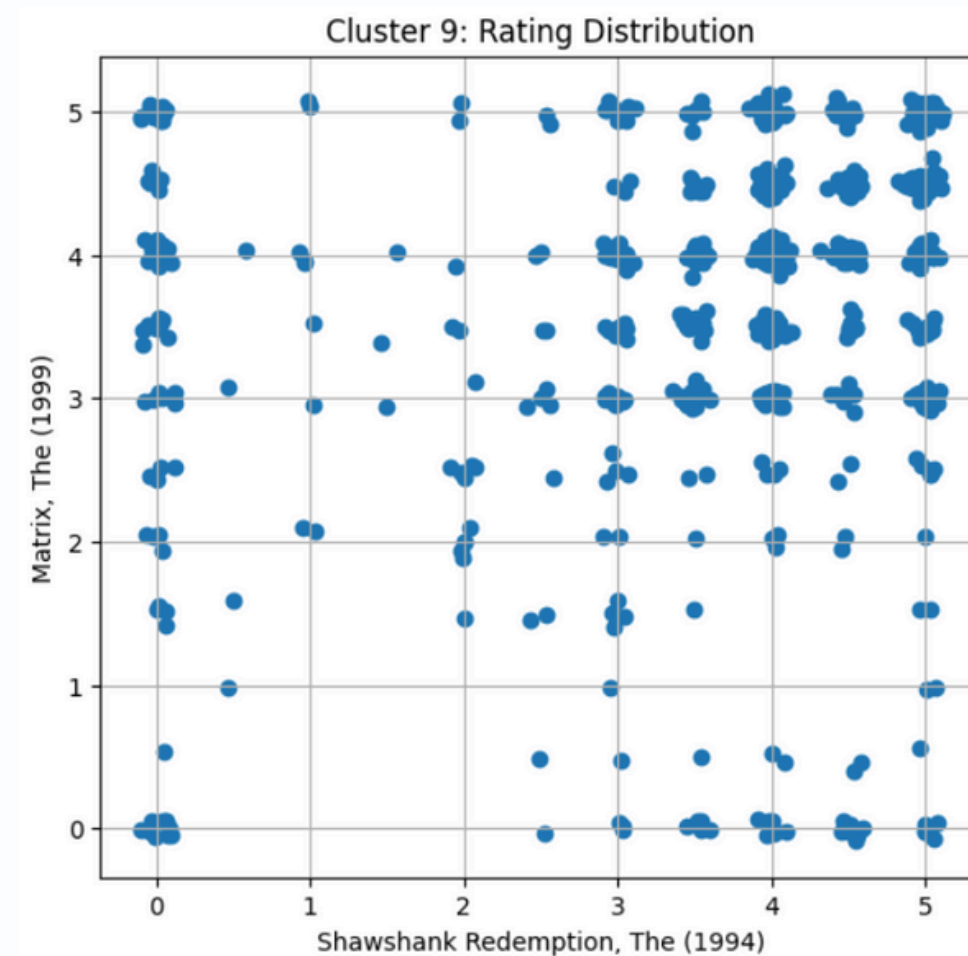


# k-Means Movies

- Selecting top 5.000 active users
- Selecting top 10.000 most rated movies
- Performing k-Means with different clusters

| userID | American Pie (1999) | '71 (2014) | Bad Moms (2016) | 'Salem's Lot (2004) | ... |
|--------|---------------------|------------|-----------------|---------------------|-----|
| 1      | 0                   | 0          | 4.5             | 4                   | ... |
| 2      | 5                   | 1          | 5               | 2                   | ... |
| 3      | 0                   | 0          | 0               | 1                   | ... |

5000 rows x 10000 columns

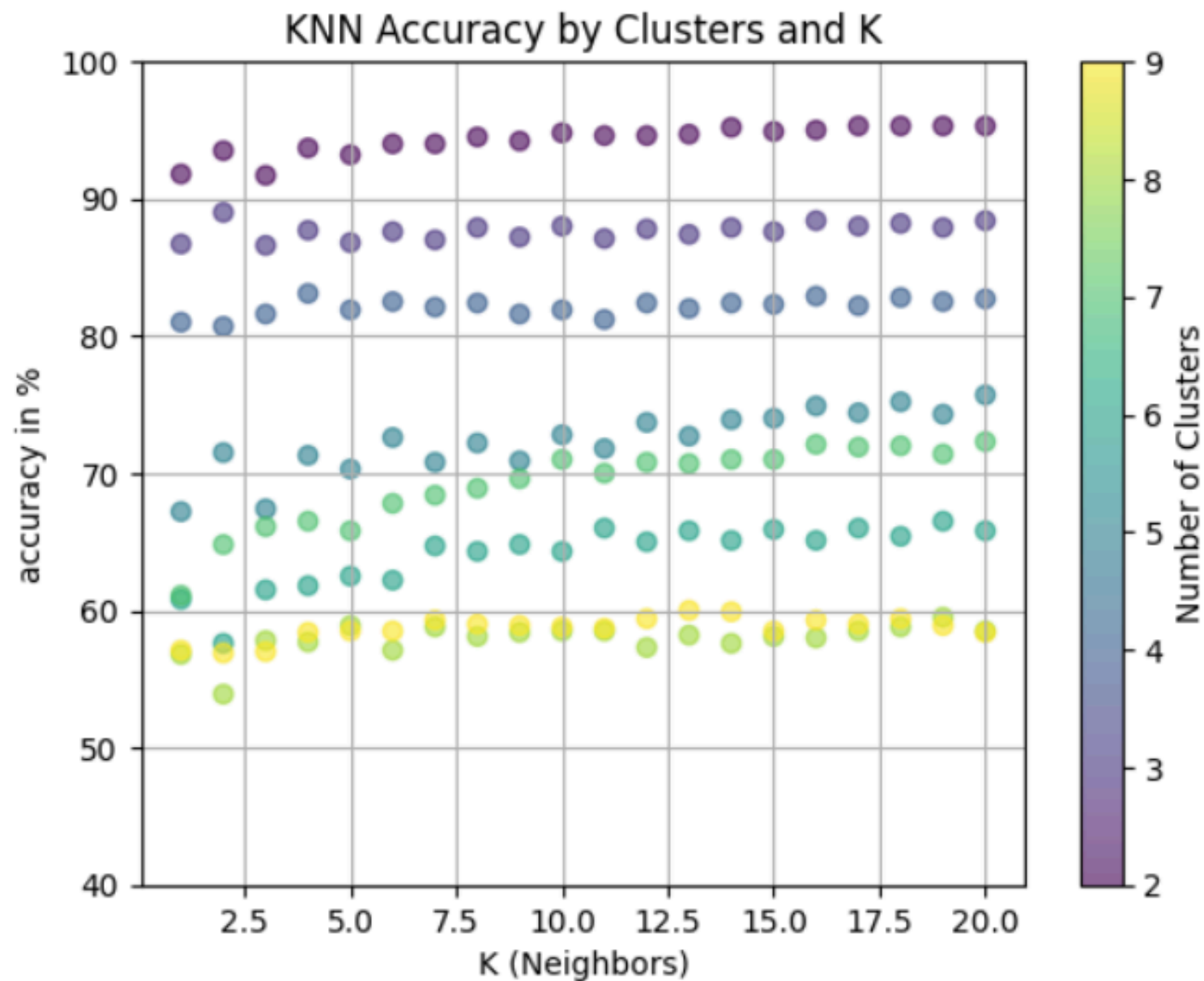


## Interpretation:

|                          | Cluster 1                     | Cluster 9                        |
|--------------------------|-------------------------------|----------------------------------|
| Rating pattern           | Concentrated, high ratings    | Dispersed, all over the spectrum |
| User preference signal   | Strong, clear                 | Weak, noisy                      |
| Recommendation potential | High (reliable taste signal)  | Low (less predictable behavior)  |
| Archetype interpretation | Fans of highly rated classics | Eclectic or unstructured viewers |



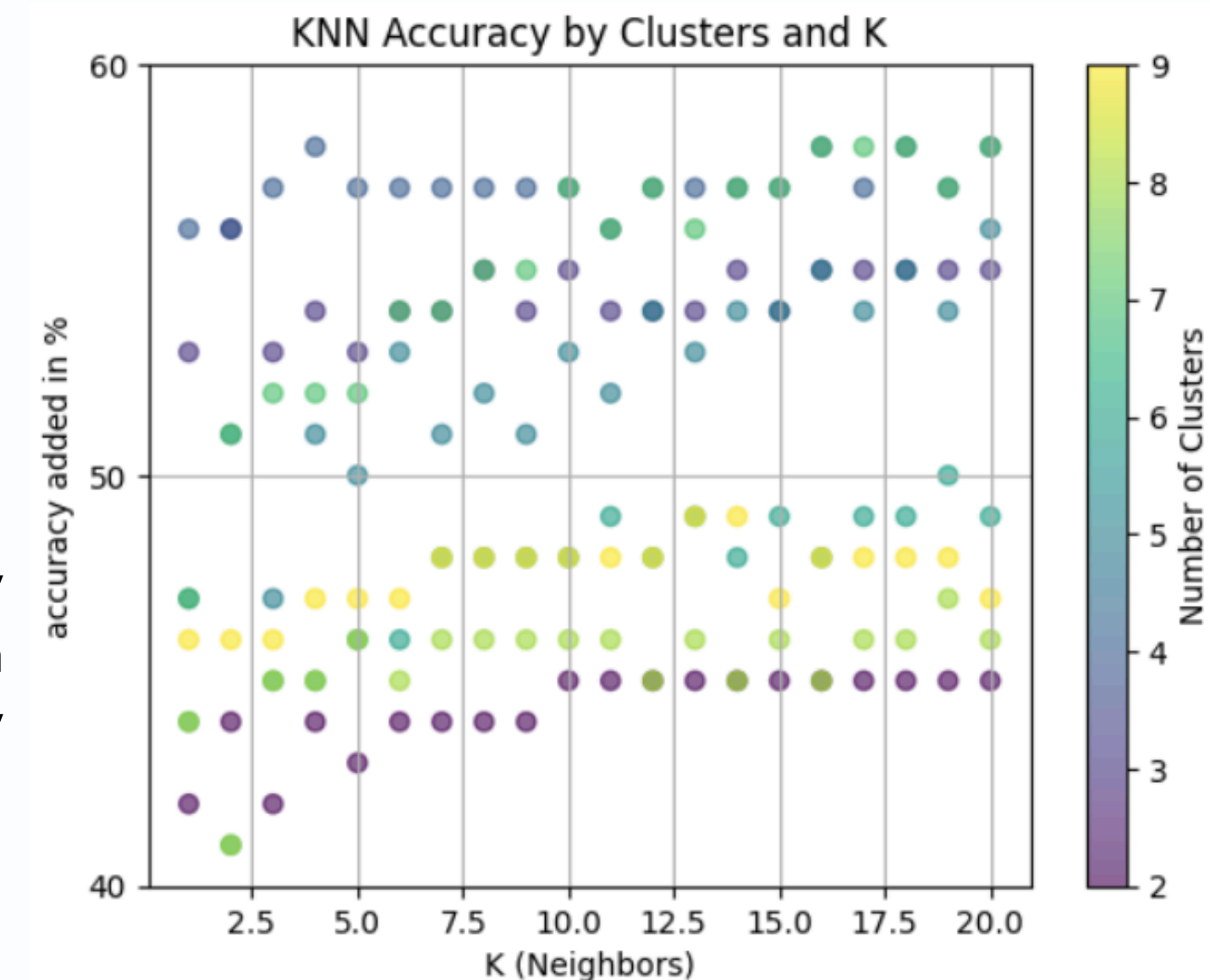
# kNN Movies Conclusion



The k-NN based on the clustered data performed as expected. The lower amount of clusters leads to higher accuracy. Therefore the configuration of 2 clusters with  $k \geq 17$  performs best.

However when looking into how much accuracy each knn adds to randomly guessing, the configuration with 4 clusters and  $k = 4$  performs best with adding 58% of accuracy

$$\text{accuracy added} = \text{accuracy} - (1/\text{amount of clusters})$$

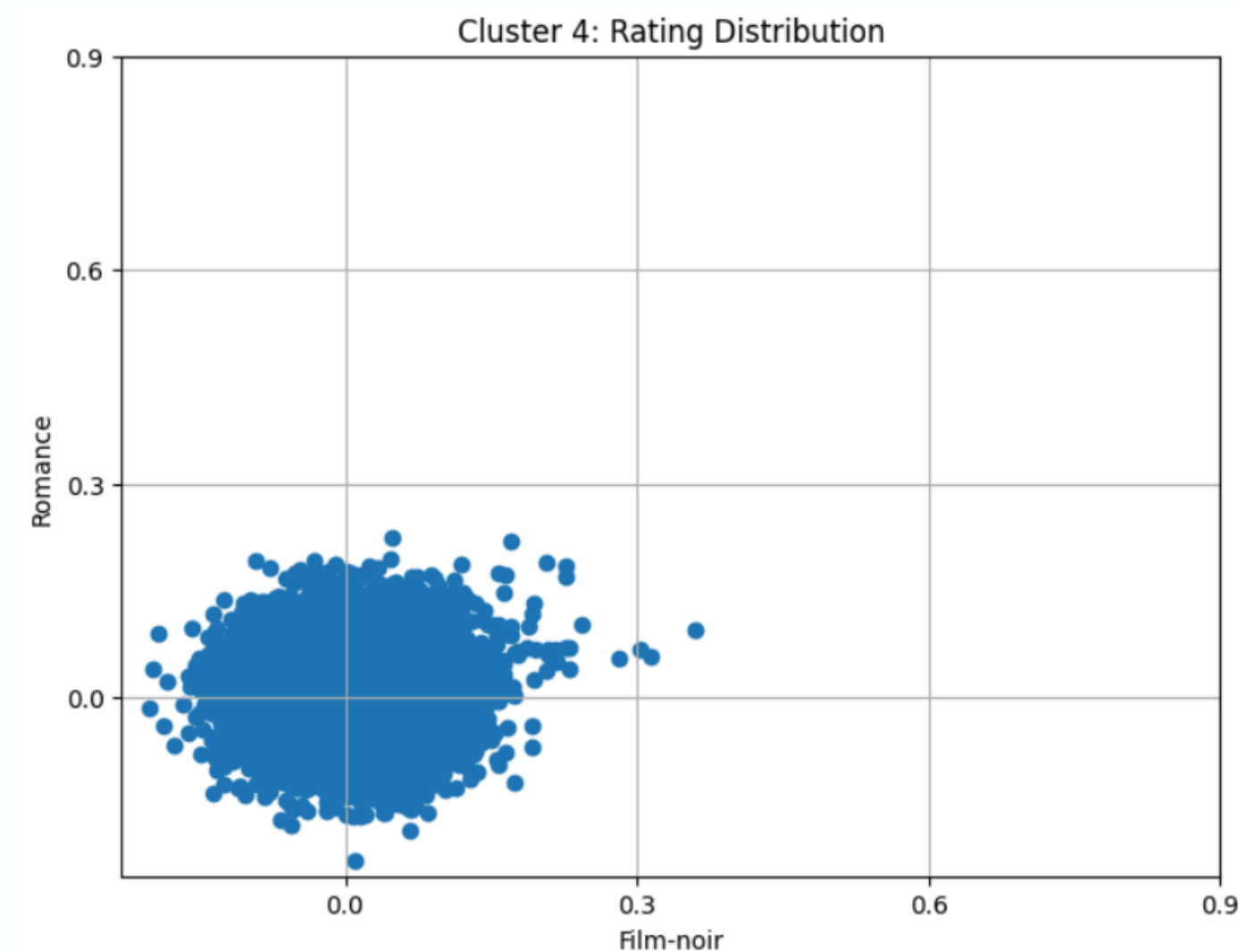
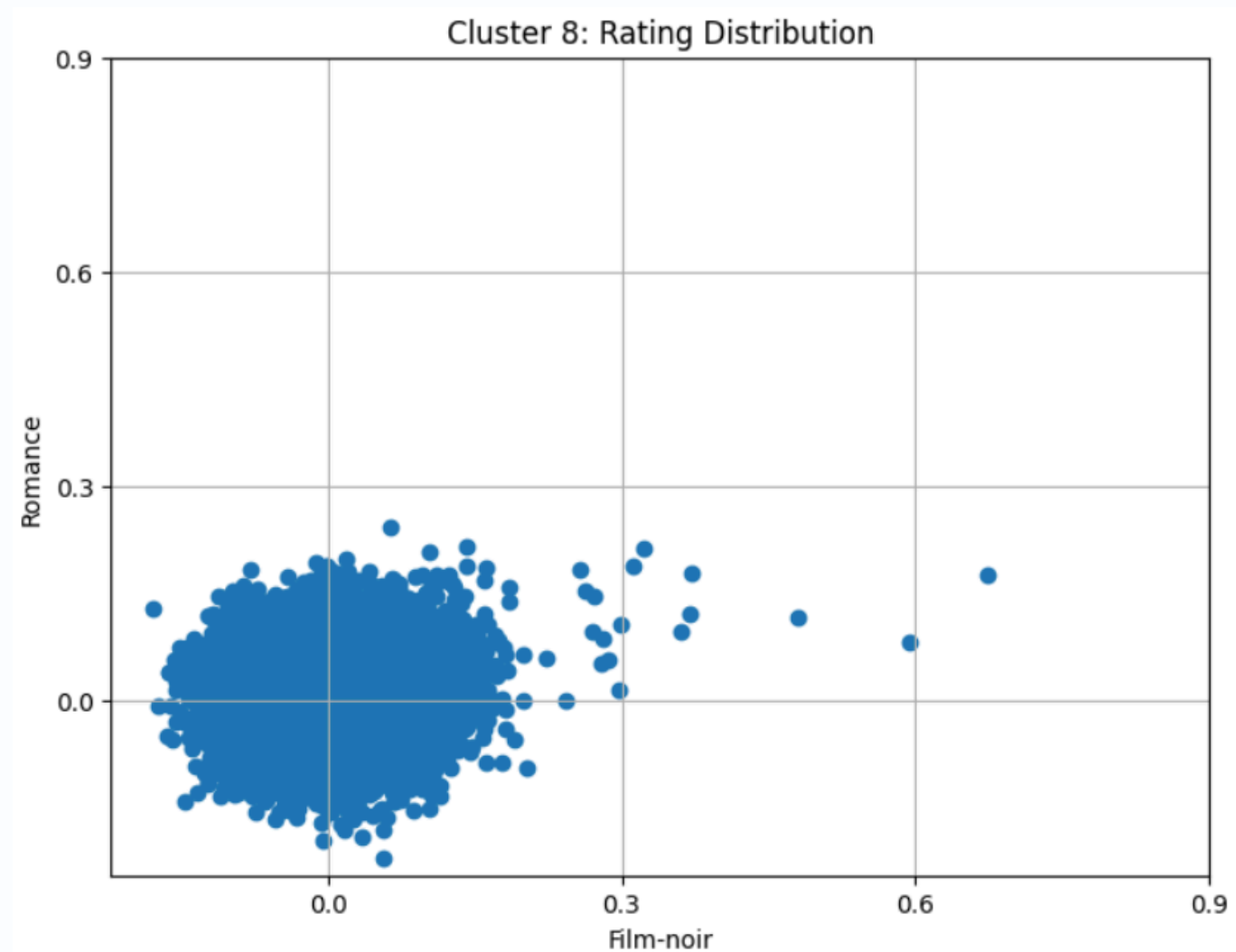


# k-Means Genres

- Splitting movies into genres
- converting each movie rating into a genre rating
- Grouping by user and summing the ratings

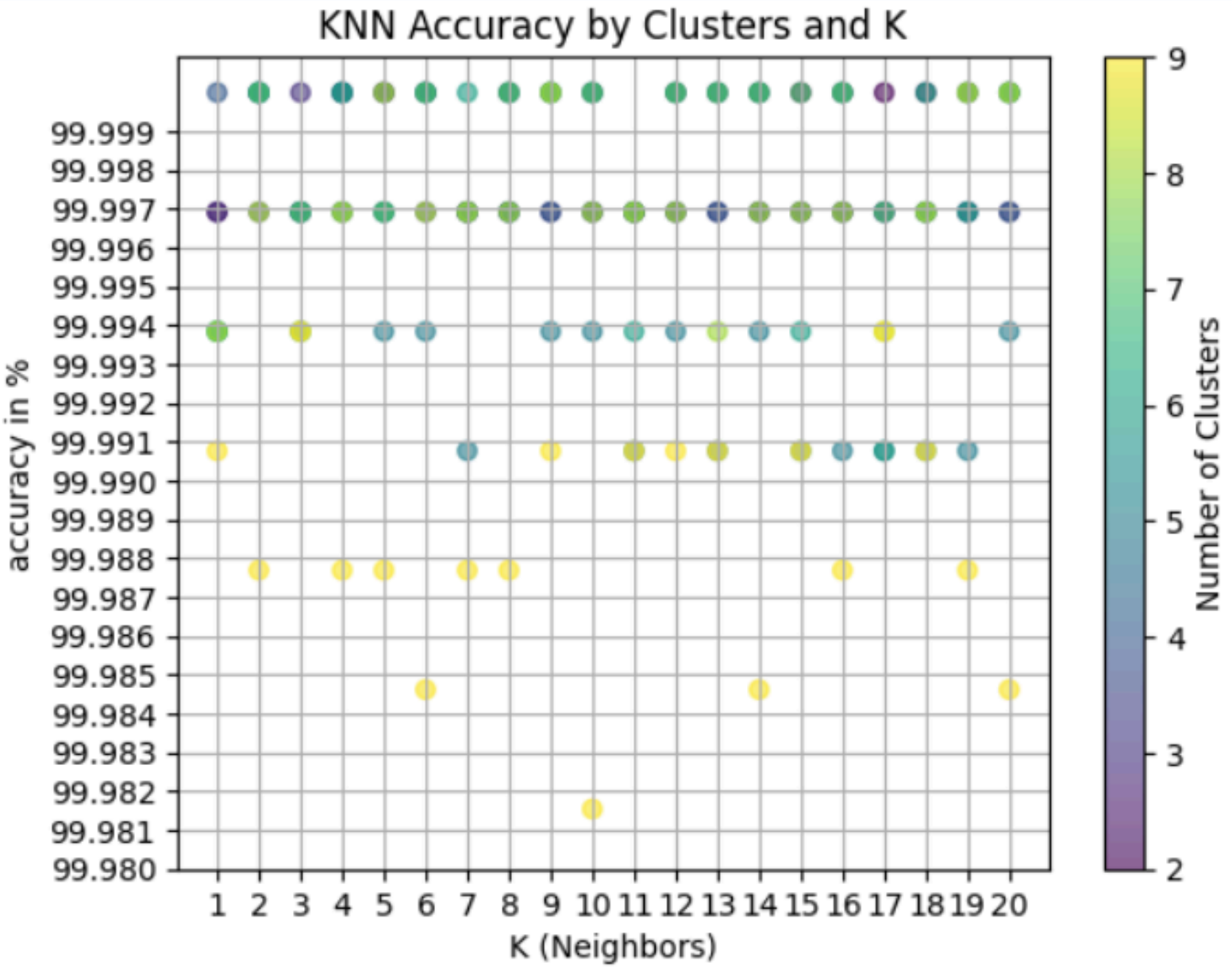
| userID | Action | Comedy | Drama | Crime | ... |
|--------|--------|--------|-------|-------|-----|
| 1      | 16.5   | 293    | 20    | 11.5  | ... |
| 2      | 302    | 121    | 44    | 32    | ... |
| 3      | 123    | 43     | 12    | 6     | ... |

162.541 rows x 21 columns



# kNN Genre

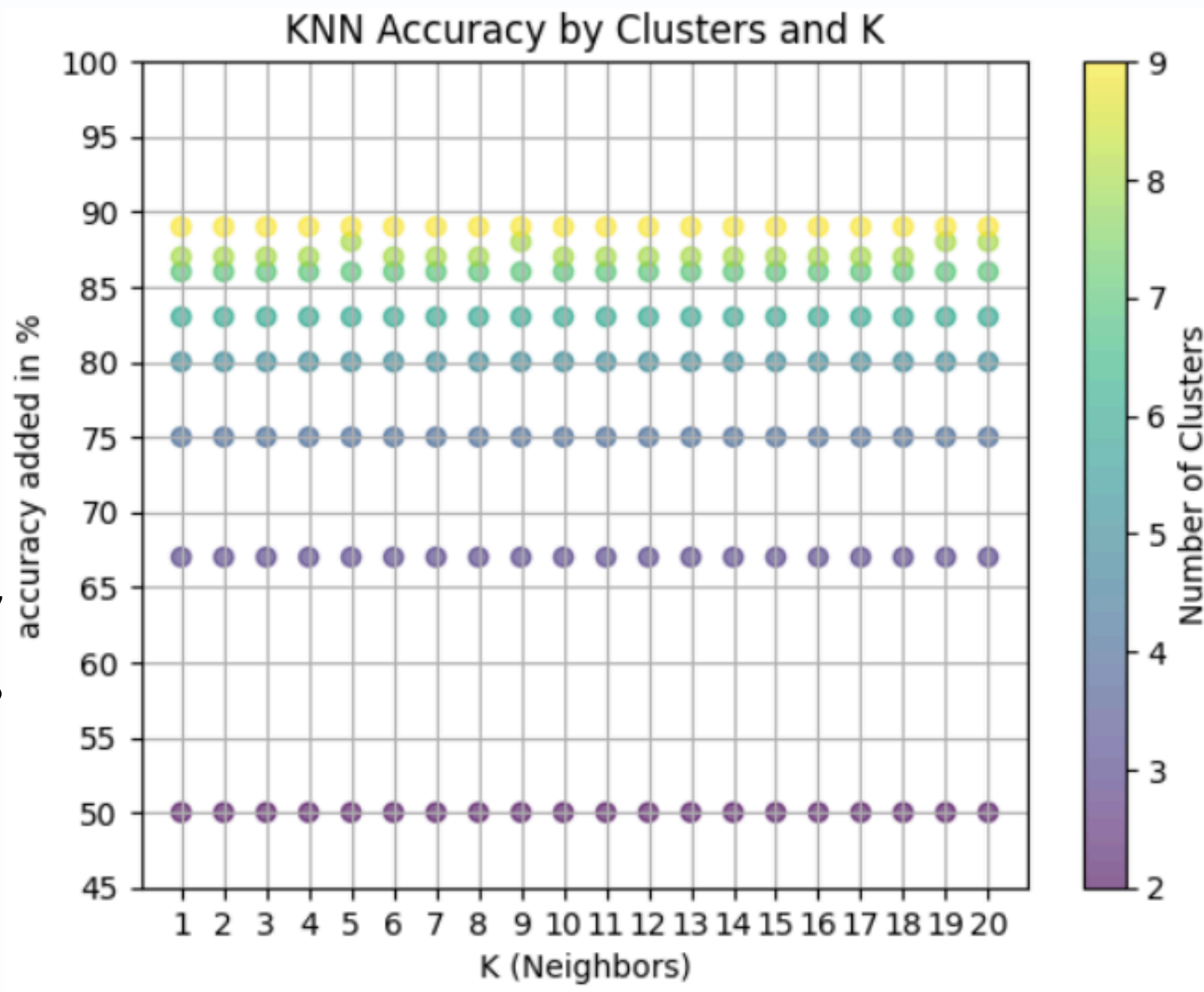
## Conclusion



The k-NN based on the clustered genre data performed as excellent.  
The data with 2 clusters reached an accuracy of 100% for almost all k values

The highest added accuracy was reached with 9 clusters and k=3 or k=17  
with an accuracy of 99% and an added accuracy of 89%

$$\text{accuracy added} = \text{accuracy} - (1/\text{amount of clusters})$$





# k-Means/kNN

Conclusion

## **CLUSTERING** EVALUATION METRICS

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### 1. MOVIE RATING-BASED CLUSTERING:

-----

- **Best Configuration: 2 clusters, K=2**
- **Classification Accuracy: 0.9030**
- **RMSE (distance to centroid): 68.0066**
- **MAE (distance to centroid): 65.6374**
- **Inertia (within-cluster sum of squares): 23124511.76**

### 2. GENRE-BASED CLUSTERING:

-----

- **Best Configuration: 9 clusters, K=8**
- **Classification Accuracy: 0.931000**
- **RMSE (distance to centroid): 0.222976**
- **MAE (distance to centroid): 0.188013**
- **Inertia (within-cluster sum of squares): 248.59**

**→ Better because  
feature space reduction  
from 10 000 dim to 20**

# k-Means/kNN

Conclusion

## RECOMMENDATION EVALUATION METRICS:

|      | Movie-Based | Genre-Based |
|------|-------------|-------------|
| RMSE | 1.1083      | 1.1128      |
| MAE  | 0.8775      | 0.8821      |

### Summary

- **Your preferences drive your cluster → Your cluster drives your recommendations.**
- **Movies are selected for their ability to differentiate tastes → Recommendations are interpretable.**
- **Validation ensures quality → Recommendations are not random, but grounded in measurable patterns.**

### Problem

- **Computational cost at training!**

**But inference → Centroid based (fast!)**



# basic recommender

recommending movies without user input

## All time classics 🏆



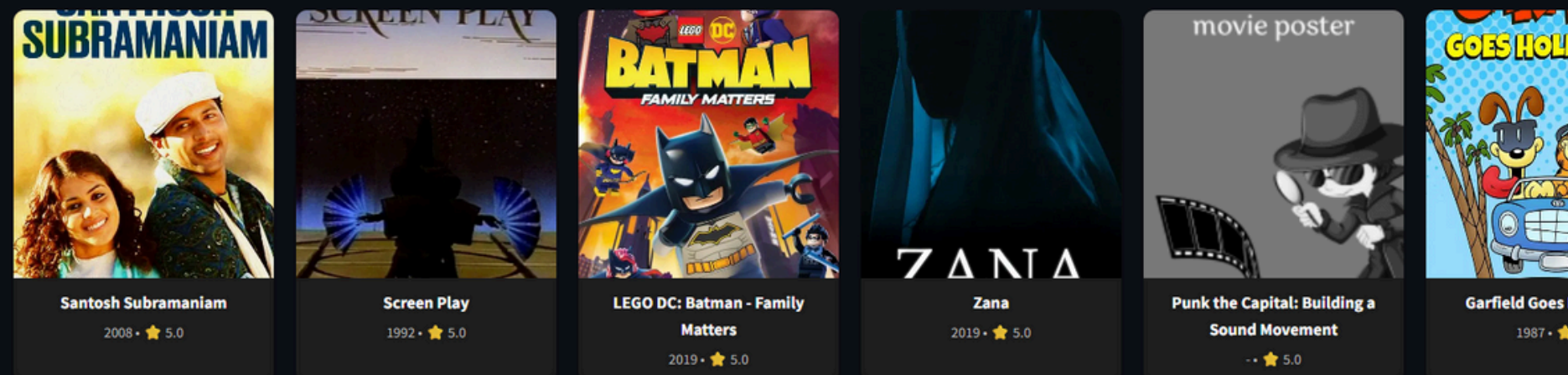
counting all ratings

grouping by genre

## Documentary



## Currently hot



adjusting ratings over time

# Results

---

| Model             | RMSE          | Strengths                            |
|-------------------|---------------|--------------------------------------|
| SVD               | 0.8           | Most accurate                        |
| NeuMF             | 1.12          | Fastest to run                       |
| k-Means/kNN       | 1.11          | Best for genre-based recommendations |
| Basic recommender | Always works! | Always works!                        |

